DRUMPANEL · MARKET AND VIABILITY

DrumPanel market and viability

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DOCUMENT CONTROL

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REVISION HISTORY

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0.1	2026-10-24	Amish Chadha	First market and viability note (kit 1.8 upgrade of a TRL 3 repo); benchmark prices checked 2026-10-25

DrumPanel market and viability

DrumPanel is for the workshops that turn used oil drums into sheet for metal art, starting with the Haitian drum artists of Croix-des-Bouquets wherever they now work, and for the cooperatives, fair trade partners and vocational schools around them. It replaces a method that costs almost nothing in money (fire, chisel, body weight) but costs smoke, noise and injuries, so its value is in safety and sheet quality, not in saving money. Commercial alternatives each do one step and cost about as much as the whole bench set: a manual drum deheader alone lists at USD 3,299, and a 36 in slip roll at USD 995 is rated below the thinnest drum wall. An open design matters because the set must be built and mended by a town fabricator. Prices were checked on the dates given. Method: STANDARDS section 23.

Table 1. Market segments.

Segment	Target users and region	Competitive benchmark (cited price)	Estimated cost per unit (USD)	Value proposition
Drum art workshops and cooperatives	Haitian metal artists (Croix-des-Bouquets and wherever they have moved), and drum artisans in other countries	Today: burning, chisels and body weight (no purchase); a ratchet-powered manual drum deheader for 55 gal drums, USD 3,299 (Global Industrial, checked 2026-10-25), which takes heads off only	3,566 in parts for one shared set (BOM), value-engineering target 3,000	Three flat 233 x 1,800 mm panels and two whole discs per drum, cold, with a written purge and vapour check
Fair trade buyers and export partners	Organisations that buy from drum artists	No documented safer preparation step found	3,566 per set, shared by a cluster	A documented, safer and repeatable first step in their supply chain, fundable as one set per cooperative
Vocational welding schools	Schools near drum art clusters	A 36 in manual slip roll, USD 995, rated for 22 gauge (0.76 mm) mild steel (Grizzly T21318, checked 2026-10-25), below the 0.8 to 1.2 mm drum wall	3,566, less if students build it	A training build in welding, turning and machine safety, and the vapour check taught as a skill
Drum collectors and recyclers	Port and industrial areas	Powered drum processing equipment (not priced here)	3,566	Cold, manual opening of oil drums that cannot be reused whole

The largest single line, the vapour check kit (USD 600), is in line with the market: a pumped-capable four-gas detector lists at USD 537 to 1,017 and a manual aspirator kit at USD 162 to 207 (Instrumart, checked 2026-10-25). A cooperative can share one detector across several workshops, but no drum is cut without it.

Open route

A town fabricator with a welder and a lathe builds the set from the drawings, the build plan and the BOM; a tool shop makes the punches, dies and discs. Fair trade partners and cooperatives carry the design to new workshops, and vocational schools can build sets as training projects. CERN-OHL-S keeps changes open, and the parametric model lets a builder change the drum size or the panel layout without a new design.

Sources

- Global Industrial, "Wizard WIZ-Kid Manual Drum Opener 13073, Non-Ferrous Cutting Wheel," USD 3,299.00, https://www.globalindustrial.com/p/wizard-174-wiz-kid-174-manual-drum-opener-13073-non-ferrous-cutting-wheel, checked 2026-10-25.
- Grizzly Industrial, "T21318 36 in Slip Roll, 22-Gauge," USD 995.00, https://www.grizzly.com/products/t21318, checked 2026-10-25.
- Instrumart, "Honeywell BW MicroClip X3 Gas Detector," USD 536.53 to 1,017.00; BW pump kit USD 162.00; D4-AS01 manual aspirator USD 207.00, https://www.instrumart.com/products/42331, checked 2026-10-25.
- SFO Museum, "Haitian Metal Sculpture," https://www.sfomuseum.org/exhibitions/haitian-metal-sculpture; Global Exchange, https://globalexchange.org/tag/haiti-oil-drum-art/ (the burning and chisel method and who does it), as cited in DMP-PRB-001.